

Zihan Hei

Statistics Undergraduate Student



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[GitHub](#)
[Quarto.pub](#)
[Personal academic website](#)

Education

AUG 2023 –
JUN 2027

Bachelor of Arts in Statistics

Binghamton University, State University of New York

Minor in Digital and Data Studies | GPA: **3.8**/4.0

Research Experience

NOV 2025 –
PRESENT

Transcriptomic Biomarker Discovery and Validation in Chemo-Immunotherapy

Independent Study with Dr. Nancy Guo | NSF REU Research Stipend | Binghamton University

- Contribute 10 hrs/week to chemo-immunotherapy biomarker discovery by reproducing RNA-seq and scRNA-seq preprocessing, clustering TCR expansion with K-means, and analyzing treatment-response signatures.
- Transform and validate single-cell RNA-seq data through quality control, normalization, gene filtering, and highly variable gene selection.
- Perform differential expression analysis, cross-study comparisons, and Gene Set Enrichment Analysis to identify and validate treatment-response biomarkers and biological pathways (manuscript in preparation).

AUG 2025 –
PRESENT

Predicting Health Information Avoidance in Colorectal Cancer

Independent Study with Dr. Heather Orom (University at Buffalo), Dr. Shane McCarty, and Dr. Kargin Vladislav | Binghamton University

- Predict health information avoidance in colorectal cancer using logistic regression, multivariate adaptive regression splines (MARS), and random forest classifiers in R with tidymodels and caret.
- Evaluate and compare model performance using cross-validation and predictive metrics, documenting results in a reproducible Quarto and LaTeX [report](#) (manuscript under review).

SEP 2025 –
APR 2026

A Mixed-Methods Analysis of Promotion and Prevention Mindsets

Undergraduate Research with Dr. Shane McCarty and 3 peers | Binghamton University

- Contributed quantitative analyses to a mixed-methods study integrating statistical modeling with team-conducted qualitative coding on community health and safety perceptions.
- Applied logistic regression modeling in R to investigate whether promotion and prevention mindsets predict health and safety-related goal preferences
- Served as first author on a poster abstract accepted for the 2026 APHA Annual Meeting and co-authored a 2025 Society for Community Research & Action presentation.

AUG 2023 –
DEC 2024

First-Year Research Immersion (FRI): Community and Global Public Health

Undergraduate Research | Binghamton University

- Collaborated with a 7-person team to research the impact of Endometriosis on women's mental health, culminating in two poster presentations to peers and faculty at FRI conferences in December 2023 and 2024.
- Conducted quantitative data analysis using SPSS for 6+ hrs/week to examine vaccine hesitancy among pregnant women.
- Designed and distributed a comprehensive campus-wide survey using Qualtrics, collecting and analyzing responses.

JUN 2025 -
PRESENT

Examining Zero-Sum Beliefs and Political Affiliation

Undergraduate Research with Dr. Shane McCarty and 2 peers | Binghamton University

- Examine political party affiliation and zero-sum beliefs as predictors of voter choice in the 2024 U.S. election.
- Analyze group differences using ANOVA/Kruskal-Wallis tests, logistic regression, and machine learning classification in R; synthesize findings into a scientific [report](#) (manuscript in preparation).

Professional Experience

JUN 2026 -
PRESENT

Quantitative Intern, First-Year Research Immersion Program

Paid Internship with Dr. Megan Fegley and Dr. Caitlin Light | Binghamton University

- Contribute 10 hrs/week of quantitative expertise to FRI program evaluation by developing R Shiny dashboards, integrating dissemination records, and designing Qualtrics surveys.
- Develop Google Apps Script automation workflows to process survey responses into spreadsheets and organize publication and presentation records.
- Visualize student outcomes and dissemination metrics in interactive R Shiny dashboards to support program evaluation and reporting.

JUN 2026 -
AUG 2026

Mental Health Workshop Quantitative Intern

Paid Summer Internship with Dr. Shane McCarty | Binghamton University

- Contributed 60 hrs of quantitative expertise to a mental health workshop by designing evaluation surveys, developing quantitative methods, and building individualized wellness profiles.
- Designed Qualtrics surveys to evaluate mental health literacy, well-being, nonviolent communication, and workshop effectiveness.
- Developed an interactive wellness webpage using Claude and Observable JS to visualize survey results and generate individualized wellness profiles.

Teaching & Mentoring Experience

JAN 2025 -
MAY 2025

R Programming Peer Mentor

First-Year Research Immersion Program | Binghamton University

- Mentored students in foundational R programming, team-based research, and Quarto reporting across public health projects.
- Collaborated with the instructor and other research assistants to support research teams' Quarto research reports

Publications

1. McCarty, S., Light, C., **Hei, Z.**, Panzik, J., Wolfe, P., & Fegley, M. (2026). *FRI Semester 3 Peer Mentor Mindsets Guide*. Binghamton University. shanemccarty.github.io/FRI/mindsets3/
2. McCarty, S., Light, C., **Hei, Z.**, & Fegley, M. (2026). *FRI Peer Mentor Mindsets Guide*. Binghamton University. shanemccarty.github.io/FRI/mindsets
3. McCarty, S., & **Hei, Z.** (2025). Chapter contributions to *The Quantitative Playbook for Public Health Research in R*, S. McCarty (Ed.).
 - McCarty, S., & **Hei, Z.** "Reproducible Reports."
 - **Hei, Z.**, & McCarty, S. "Import Data Once."
 - **Hei, Z.**, & McCarty, S. "Read Me."
 - **Hei, Z.** "Writing Methods & Results Reproducibly."
 - **Hei, Z.** "APA Citations & References."
 - **Hei, Z.** "Publish Your Report to Quarto Pub."shanemccarty.github.io/FRI/playbook/

Presentations & Conference Abstracts

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| 2026 | A Mixed-Methods Analysis of Promotion and Prevention Mindsets
<i>American Public Health Association Annual (APHA) Meeting</i> <ul style="list-style-type: none">• Hei, Z., Giallella, J., John, J., Khalap, I., Booker, J., Pacque, K., Rualo, G. & McCarty, S. First-author poster abstract accepted for presentation.
https://apha.confex.com/apha/2026/meetingapp.cgi/Paper/594471 |
| NOV 2025 | Promotion & Prevention Mindsets: A Four-Study, Mixed-Methods Approach
<i>Society for Community Research & Action (online)</i> <ul style="list-style-type: none">• Invited presentation; McCarty, S., Hei, Z., Rualo, G., Khalap, I., & Giallella, J. |
| DEC 2024 | Public Knowledge & Attitudes on Vaccinations and Reproductive Health
<i>FRI Proposal Poster Research Session 11th annual, Binghamton University</i> |
| DEC 2023 | Endometriosis and Women's Mental Health
<i>FRI Proposal Poster Research Session 10th annual, Binghamton University</i> |

Data Science Projects

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| JAN 2026 -
PRESENT | Histopathology Thyroid Cancer Detection Using YOLO26 Group Project
<i>Digital & Data Studies Group Project Binghamton University</i> <ul style="list-style-type: none">• Develop a YOLO26-based deep-learning workflow for thyroid cancer, abnormal, and normal histopathology tissue detection, including model training, evaluation, and error analysis.• Evaluate model performance through Type I (false positive) and Type II (false negative) error analysis and interpret confusion matrices for class imbalance and dataset representativeness.• Visualize and document model performance and limitations for potential pre-screening applications (manuscript in preparation). |
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SEP 2025 -
DEC 2025

Deep Reinforcement Learning in Board Games | [Group Project](#)
Digital & Data Studies Group Project | Binghamton University

- Implemented PyTorch reinforcement-learning agents for Tic-Tac-Toe and Connect Four through self-play and visualized training performance using Matplotlib.

Leadership & Activities

SEP 2026 -
PRESENT

Graphic Designer
Binghamton AI Society, Binghamton University

- Design the club logo and promotional materials, including event posters, workshop graphics, and outreach visuals.

Awards, Honors & Scholarships

FEB 2026 -
FEB 2027

NSF Research Experiences for Undergraduates (REU) Research Stipend Recipient
Binghamton University

- Awarded an \$8,000 year-long NSF REU research stipend supporting transcriptomic biomarker discovery and validation in chemo-immunotherapy with Dr. Nancy Guo.

SUMMER 2026

Fleishman Internship Funding Award
Binghamton University

- Awarded \$3,000 in summer internship funding to support a quantitative research internship with the First-year Research Immersion program.

FEB 2026

Pi Mu Epsilon
National Mathematics Honor Society

FALL 2023 -
PRESENT

Harpur College Dean's List
Binghamton University

- Named to the Dean's List every semester attended.

EACH
SEMESTER

Provost's Promise Scholarship
Binghamton University

- Recipient of renewable financial support each semester throughout undergraduate study.

Technical Skills

Programming: R (tidyverse, tidymodels, caret), Python (Pandas, NumPy, scikit-learn, PyTorch), SPSS

Data Visualization & Applications: ggplot2, Plotly, Matplotlib, Observable JS, R Shiny, Qualtrics

Research & Reproducibility: Quarto, GitHub, LaTeX

Methods: Regression, random forests, decision trees, MARS, deep learning, differential expression, GSEA, mixed-methods analysis

Communication & Collaboration: Public speaking; Team-based research in academic and internship settings
